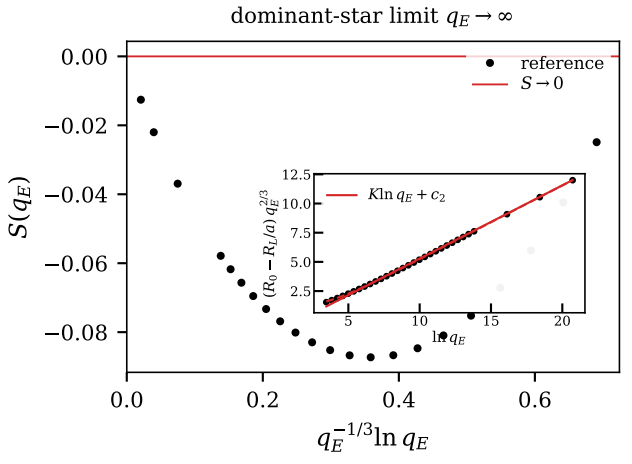
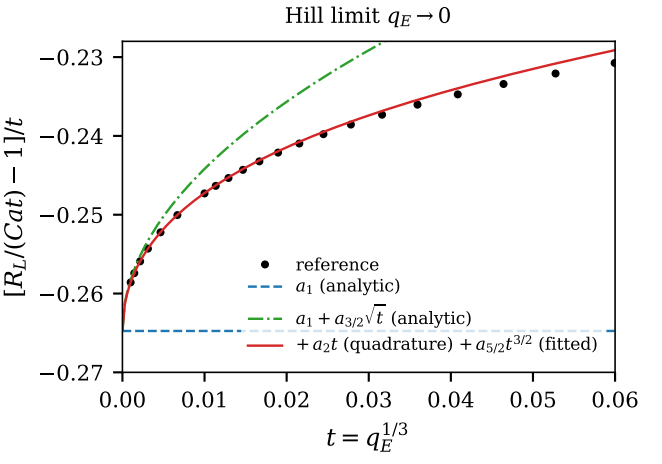

dominant-star limit $q_E \to \infty$
reference
$S \to 0$
$S(q_E)$
$q_E^{-1/3} \ln q_E$
$K \ln q_E + c_2$
$(R_0 - R_L(a))\, q_E^{2/3}$
$\ln q_E$


Hill limit $q_E \to 0$
reference
a_1 (analytic)
$a_1 + a_{3/2}\sqrt{t}$ (analytic)
$+ a_2 t$ (quadrature) $+ a_{5/2} t^{3/2}$ (fitted)
$[R_L/(Cat) - 1]/t$
$t = q_E^{1/3}$